

Notes on Representation Theory

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Representation theory is an immense topic. Most sources we have encountered emphasize rigor and depth over conceptual clarity, especially for physics. Such presentations tend to conceal the exceptional regularity and beauty of the subject just due to their length. Here, we present the bare minimum, from the physicist's perspective, following parts of Schwichtenberg¹ closely. In addition to resolving ambiguities as we find them, we specifically bridge classical physics, where representation theory is not often invoked, but where we find it quite illuminating, and quantum physics, where the physics student typically first encounters the subject. We consider this unfortunate, because the student both misses the clarifying utility of representation theory in classical physics and because the student of quantum physics already has many steep hills to climb. We deem it better to cover representation theory as a separate topic pertinent to both realms of physics.

¹ Jakob Schwichtenberg. *Physics from Symmetry*. Springer Nature, 2nd edition, 2018. ISBN 978-3-319-66630-3

Notation

We employ Schwichtenberg's variations of the equality sign:

Symbol	Meaning
$\equiv$	Is defined as
$\triangleq$	Corresponds to / is represented by
$\stackrel{!}{=}$	Is required to be / must hold that

Prerequisites

Arfken² is excellent background for these notes. Look there for undefined terms like *vector space* or *differentiable manifold*. We adhere to the following terse definitions in the rest of these notes.

² George B. Arfken, Hans J. Weber, and Frank E. Harris. *Mathematical Methods for Physicists*. Academic Press, 7th edition, 2012. ISBN 978-0123846549

Definition 1 (Lie Algebra). A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anticommutative, binary operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ that satisfies the Jacobi identity:

$$[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$$

Definition 2 (Lie Group). A Lie group is a group and also a differentiable manifold.

It bears emphasis that a Lie algebra is not a group under the Lie bracket and is not a differentiable manifold.

Generators

The rest of this document concerns the relationships between Lie groups and Lie algebras. The first of these relationships comes via *generators*.

Definition 3 (Generator). A generator $X \in \mathfrak{g}$ is a Lie-algebra element that, multiplied by a real θ ,³ produces an element of a Lie group G under the exponential map:

$$\begin{aligned} \exp : (\mathfrak{g} \times \mathbb{R}) &\rightarrow G \\ h(\theta) &= \exp(\theta X) \end{aligned}$$

The exponential map is abstract, here. It must be further defined for any concretization. Physicists work almost exclusively with matrix concretizations for the sake of calculation. In such concretizations, $\exp$ is a matrix exponential, defined by Taylor series. From this point on, assume all concretizations are matrix concretizations. We hope it's obvious from context when we're working abstractly and when we're working concretely.

Remark 1. A generator $X \in \mathfrak{su}(2)$ in the Lie algebra can be recovered from a Lie-group element $h(\theta)$ as follows:

$$X = \left. \frac{dh(\theta)}{d\theta} \right|_{\theta=0}$$

Many Lie groups can map to the same Lie algebra.

Example 1. $SO(3)$ and $SU(2)$ have the same Lie algebra: $\mathfrak{su}(2)$. Begin by considering $SO(3)$ ⁴ as a matrix group: the set of real, 3×3 orthogonal matrices with determinant 1.

Proposition 1. Real 3×3 generators $J \in \mathfrak{su}(2)$ of the special orthogonal group $SO(3)$ are traceless and antisymmetric.

Proof.

- *Traceless:* “Special” means that the elements $O \in SO(3)$ have unit determinant. That fact implies that the generators are traceless:⁵

$$\det(O) \stackrel{!}{=} 1 \quad \implies \quad \det(e^{\theta J}) = e^{\theta \text{Tr}(J)} \stackrel{!}{=} 1 \quad (\text{Sch:3.60})$$

- *Antisymmetric:* “Orthogonal” means that elements O of $SO(3)$ satisfy $O^T O = O O^T \stackrel{!}{=} 1$. Therefore:

$$O^T O = e^{\theta J^T} e^{\theta J} \stackrel{!}{=} 1 \quad \implies \quad J^T \stackrel{!}{=} -J \quad (\text{Sch:3.59})$$

The *transpose* operation lifts into the exponent, as can be seen by a Taylor expansion, because θ is a scalar.

³ X is a vector in the Lie Algebra $\mathfrak{g}$ as a vector space, and vectors can always be multiplied by scalars from the underlying field of the vector space, by vector-space rules

⁴ $SO(3)$ is also a Lie group, of course

⁵ Equation numbers with prefixes, such as (Sch:3.60), are taken from other sources that are meant to be obvious from the prefix.

□

Both $\mathfrak{su}(2)$ and $\mathrm{SO}(3)$ are three-dimensional, $\mathfrak{su}(2)$ in the sense that a basis requires three linearly independent generators $\{X_1, X_2, X_3\}$ to span the space, $\mathrm{SO}(3)$ in the sense that its matrix form consists of 3×3 matrices. A basis for the following *adjoint representation*^{6,7} of $\mathfrak{su}(2)$ is

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (\text{Sch:3.61})$$

We note, in passing, that the Lie bracket can “generate” the basis (in a different sense of the word *generate*), from just two of the J vectors / matrices,⁸ because $[J_1, J_2] = J_3$, etc.

The following Mathematica code checks that these generators in $\mathfrak{su}(2)$ satisfy the Lie-algebra commutation relations $[J_i, J_j] = \epsilon_{ijk} J_k$.⁹

```

Mathematica
J1 = {{0, 0, 0}, {0, 0, -1}, {0, 1, 0}};
J2 = {{0, 0, 1}, {0, 0, 0}, {-1, 0, 0}};
J3 = {{0, -1, 0}, {1, 0, 0}, {0, 0, 0}};

J1.J2 - J2.J1 == J3 (* True *)
J2.J3 - J3.J2 == J1 (* True *)
J3.J1 - J1.J3 == J2 (* True *)

```

Mathematica can further check that the matrix exponentials of these generators span $\mathrm{SO}(3)$ as a matrix group. The following should suffice to convince you of this fact, assuming that you are familiar with elementary rotation matrices:

```

Mathematica
MatrixExp[\[Theta] J1]
{{1, 0, 0},
 {0, Cos[\[Theta]], -Sin[\[Theta]]},
 {0, Sin[\[Theta]], Cos[\[Theta]]}}

```

*Proposition 2. Complex 2×2 generators $T \in \mathfrak{su}(2)$ of the special unitary group $\mathrm{SU}(2)$ are traceless and anti-Hermitian.*¹⁰

Proof. Proof of the traceless property goes just as above.

“Unitary” means that the elements U of $\mathrm{SU}(2)$ satisfy $U^\dagger U = U U^\dagger \stackrel{!}{=} 1$, where $U^\dagger$ is complex conjugate transpose, or *adjoint*, symbolized by the $\dagger$.¹¹

“Hermitian” means that a matrix is self-adjoint. “Anti-Hermitian” means that a matrix equals the negative of its adjoint. Now we see that

$$U^\dagger U = e^{\theta^\dagger} e^{\theta} \stackrel{!}{=} 1 \implies T^\dagger \stackrel{!}{=} -T \quad (1)$$

⁶ This usage of “adjoint” is distinct from the adjoint operation on matrices and is explained in the next section.

⁷ Informally, for now, take “representation” to mean “matrices corresponding to elements of a Lie group or Lie algebra.” We’ll make this notion rigorous shortly.

⁸ “Vectors” in $\mathfrak{su}(2)$ are matrices in this representation.

⁹ ϵ_{ijk} is the totally antisymmetric Levi-Civita symbol that equals 1 when ijk is an even permutation of 123, -1 when ijk is an odd permutation, and 0 otherwise. This expression is *not* using the Einstein summation convention; it’s just ordinary scalar-vector (as matrix) multiplication for any i, j , and k .

¹⁰ “Anti-Hermitian” for complex matrices is analogous to antisymmetric for real matrices like the adjoint representation of $\mathrm{SO}(3)$ above.

¹¹ As noted above, this meaning of “adjoint” is distinct from its meaning in “adjoint representation”, which pertains to the representation of $\mathrm{SO}(3)$ above.

The adjoint operation also lifts into the exponent, like *transpose*, because θ is a *real* scalar.

□

A basis for this *fundamental representation*¹² is $-i/2$ times the three *Pauli matrices*:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{Sch:3.80})$$

The basis is $T_1 = -i\sigma_1/2$, $T_2 = -i\sigma_2/2$, $T_3 = -i\sigma_3/2$ as can be readily verified in Mathematica, showing, again in passing, that the factor of $1/2$ maintains the unit *structure constants* ϵ_{ijk} .¹³ Without the $1/2$, we'd be chasing around factors of 2. This fact is related to the *double covering* of the fundamental representation over the adjoint representation, a notion we'll get back to.

¹² Clarified below.

¹³ We'll see more diverse structure constants in the sequel.

Mathematica

```
T1 = -I/2 PauliMatrix[1];
T2 = -I/2 PauliMatrix[2];
T3 = -I/2 PauliMatrix[3];
T1 . T2 - T2 . T1 == T3 (* True *)
T2 . T3 - T3 . T2 == T1 (* True *)
T3 . T1 - T1 . T3 == T2 (* True *)
```

Representations

Representations are maps from a Lie group or a Lie Algebra to a vector space of matrices. The representations of a Lie group preserve the group properties and the representations of a Lie algebra preserve the Lie bracket. Formally:

Definition 4 (Lie Group Representation). A representation of a Lie group G is a map Π from any Lie group element $g \in G$ to a linear transformation $\Pi(g)$ of some vector space V such that group properties are preserved.

$$g \xrightarrow[\Pi]{} \Pi(g) \quad (\text{Sch:3.65})$$

Remark 2. Preserving group properties means:

Multiplication is preserved: $\Pi(g_1 g_2) = \Pi(g_1) \Pi(g_2)$

Group identity maps to matrix identity: $\Pi(e) = I$

Inverses are preserved: $\Pi(g^{-1}) = \Pi(g)^{-1}$

Definition 5 (Lie Algebra Representation). A representation of a Lie algebra $\mathfrak{g}$ is a map π from any Lie algebra element $X \in \mathfrak{g}$ to a linear

transformation $\pi(X)$ of some vector space V such that the Lie bracket is preserved:

$$\pi([X, Y]) = [\pi(X), \pi(Y)] \quad (2)$$

where the bracket on the right is the standard matrix commutator:
 $[\pi(X), \pi(Y)] \equiv \pi(X)\pi(Y) - \pi(Y)\pi(X)$.

When a Lie group and a Lie algebra are linked by the exponential map, every representation of a group “lifts” to a representation of its algebra, and vice versa. Figure 1 shows the relationships amongst:

The Lie algebra $\mathfrak{g}$: e.g., $\mathfrak{su}(2)$, spanned by T in the fundamental and by J in the adjoint representation

The Lie group G : e.g., $G_{\text{sc}} = \text{SU}(2)$, the unique simply connected group, or $G = \text{SO}(3)$, the multiply connected group of rotation matrices in $\mathbb{R}^3$

The Carrier Space V : e.g., for $\text{SU}(2)$, $V = \mathbb{C}^2$, the *spinor space*; for $\text{SO}(3)$, $V = \mathbb{R}^3$, the physical Euclidean space

The General Linear Group $\text{GL}(V)$: the group of all invertible linear transformations of the carrier space

The General Linear Algebra $\mathfrak{gl}(V)$: the space of all linear maps from V to V that preserve the Lie bracket as the matrix commutator
 $[A, B] = AB - BA$

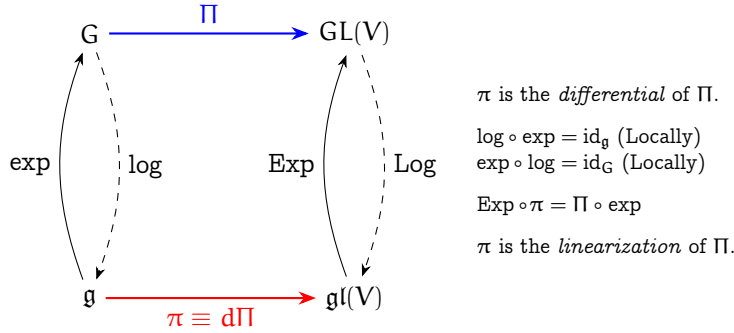


Figure 1: The relationship amongst Lie group and Lie algebra representations.

The Two Worlds of $\mathfrak{su}(2)$

In our motivating example of $\mathfrak{su}(2)$, we found two distinct bases: a 2×2 complex basis $\{T_a\}$ and a 3×3 real basis $\{J_a\}$. To a physicist, these are not just different sizes of matrices; they represent two different ways a group can act on the physical world.

The Fundamental Pair (Direct Action)

The 2×2 representation is the *fundamental representation*. It describes how the group $SU(2)$ acts on its “native” carrier space $V = \mathbb{C}^2$ (the space of spinors).

- **Group Level:** $\Pi_{\text{fund}}(Q)$ is the 2×2 matrix itself.
- **Algebra Level:** $\pi_{\text{fund}}(X)$ is the 2×2 generator.
- **Action:** The *direct action* $Q|\xi\rangle$, where $Q \in SU(2)$ and $|\xi\rangle$ is a spinor, produces a new spinor.

The Adjoint Pair (Adjoint Action)

The 3×3 representation is the *adjoint representation*. It describes how the group $SU(2)$ acts on its own Lie algebra $\mathfrak{su}(2)$.

- **Group Level:** $\Pi_{\text{adj}}(Q)$ is the 3×3 rotation matrix in $SO(3)$.
- **Algebra Level:** $\pi_{\text{adj}}(X)$ is the 3×3 matrix of structure constants $(f_a)_{bc} = \epsilon_{abc}$.
- **Action:** The *adjoint action* $QXQ^\dagger$, where $Q \in SU(2)$ and X is a vector in the algebra, produces a new vector in the algebra.

Remark 3. Note the “squaring” relationship: The adjoint representation is (roughly) the product of two fundamental representations ($Q \dots Q^\dagger$). This is why $SO(3)$ is the adjoint group of $SU(2)$. It is also why spinors ($SU(2)$) are the “square root” of vectors ($SO(3)$), and why it takes a 4π rotation to return a spinor to its original state, while 2π suffices for a vector. Another view is that the relationship between the fundamental and adjoint representations explains the *double covering* of $SO(3)$ by $SU(2)$.

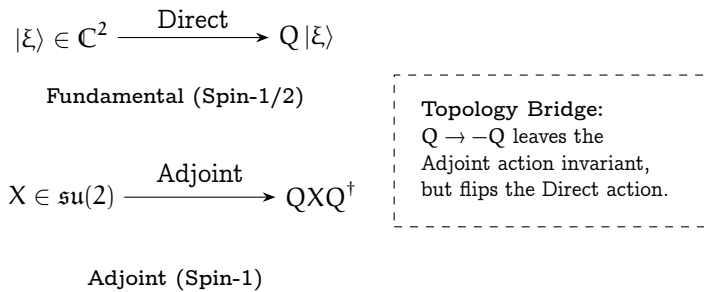


Figure 2: Direct vs. Adjoint actions of $SU(2)$.

Fact 1. *The Adjoint representation is always available for any Lie group. It is the “regulation” way to view a Lie group as a matrix group of rotations on its own algebra.*

References

George B. Arfken, Hans J. Weber, and Frank E. Harris. *Mathematical Methods for Physicists*. Academic Press, 7th edition, 2012. ISBN 978-0123846549.

Jakob Schwichtenberg. *Physics from Symmetry*. Springer Nature, 2nd edition, 2018. ISBN 978-3-319-66630-3.